Skylar: Troubleshooting Skylar Backend

This Guide is for troubleshooting the infrastructure of Skylar. Application issues are covered in other guides.

Guide	Skylar
Tags	DevOps, install, installation, Skylar, SL1 setup, Skylar connection
Team Owner	Skylar
Status	ACTIVE
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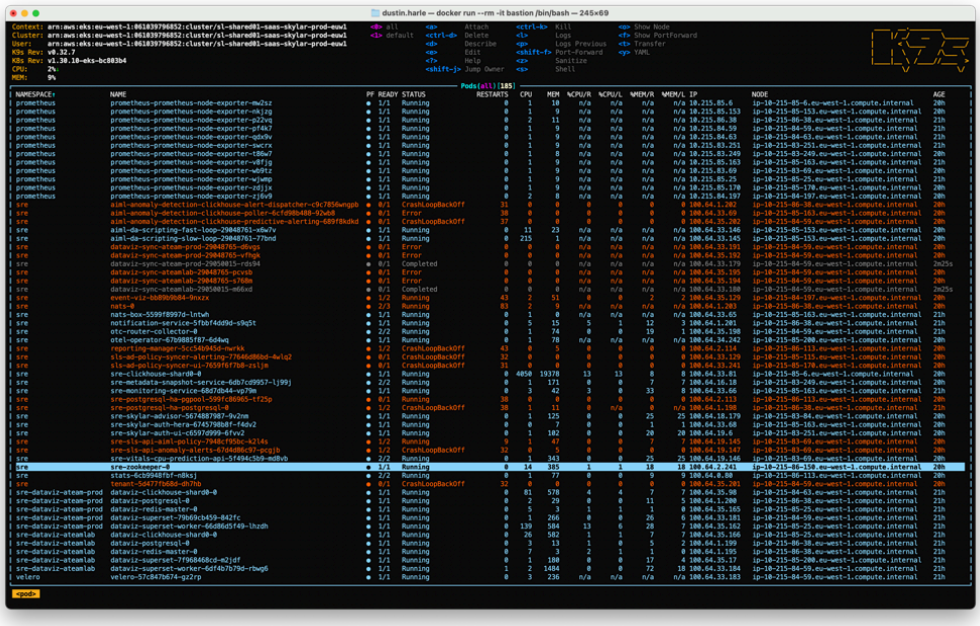
Multiple Pods crashing:

You login to the cluster and notice a large majority of pods crashing like the screenshot below. Where do you start?

Check the core services pods in the table below.

Service	Services	What to check?
Clickhouse:	beta3-clickhouse-shard0-0 beta3-zookeeper-0	Pod Status Disk space
PostGres	beta3-postgresql-ha-postgresql-0 beta3-postgresql-ha-pgpool	Pod Status Disk Space

In this example screenshot all pods are crashing due to PostGres pod being unstable.



HTTP Error codes map:

If clients start to receive an HTTP response other than 200.

HTTP Response	Service to check	Possible Reason
502	otc-router-collector-0	Check router logs.
401	Tenant-service	Tenant service is unable to validate API key either because it's wrong or service is broken (ie: connection to postgres to validate token). You can refer to Skylar cluster monitoring | Opentelemetry which has metric

		`otelcol_mgmt_last_date`. Check this metric to see when the OTEL config was last changed.
403	ingress-nginx	Most likely the client IP is not in the ingress whitelist. Reach out to SRE/Engineering to have the DataEngine egress IPs added.

Check Nginx pod logs

```
neil.toor@Toor-Neils-MacBook-Pro advisor % kubectl logs nginx-ingress-controller-internet-facing-57b9866899-hnf25 -n nginx | grep " 403 "
52.91.183.94 - - [03/Apr/2025:16:38:27 +0000] "GET /bitbucket/.git/config HTTP/1.1" 403 146 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/17.5 Safari/605.6.18" 296 0.000 [beta3-dataviz-fastlane-default-dataviz-clickhouse-8123] [] - - - - 03958465d5c751d7667dbba23dcee684
```

Somethings not right with the UI:

Check pod logs for below services

```
beta3-skylar-auth-hera-5fd5756df7-t8x44
beta3-skylar-auth-ui-67ff765fb8-7dzjk
```

DNS Issues:

DNS is handled by external-dns pod running in kube-system namespace. For any DNS issues take a look at the pod logs. The most common issue is auth to the Rout53 Hosted Zone.

External-DNS can create records either by using an [IAM roles]() or AWS access keys. To see auth mapping take a look at the external-dns Service Account.

```
kubectl edit sa external-dns -n kube-system
```

Certificate Issues:

Certificates are automatically created/rotated via cert-manager. Take a look at cert-manager controller logs inside the cert-manager namespace.

You can also check the issuance of certificates by checking the below CRD objects.

```
neil.toor@Toor-Neils-MacBook-Pro advisor % kubectl get certificate -A
NAMESPACE                                NAME
READY   SECRET                                                                  AGE
beta3-dataviz-accenture-default          cert-tls
True    cert-tls                                                                29d
beta3-dataviz-accenture-default          cert-tls-odbc
True    cert-tls-odbc                                                           29d
beta3-dataviz-bensuurmeijer-instance     cert-tls
True    cert-tls                                                                75d


neil.toor@Toor-Neils-MacBook-Pro advisor % kubectl get certificaterequest
NAME                            APPROVED   DENIED   READY   ISSUER                REQUESTOR
AGE
cert-tls-1                      True                True    letsencrypt-route53
system:serviceaccount:cert-manager:cert-manager-controller   13d
cert-tls-clickhouse-ext-new-1   True                True    letsencrypt-route53
system:serviceaccount:cert-manager:cert-manager-controller   3d1h

neil.toor@Toor-Neils-MacBook-Pro advisor % kubectl get order -A
NAMESPACE                                NAME                                      STATE
AGE
beta3-dataviz-accenture-default          cert-tls-1-2627459209                     valid
29d
beta3-dataviz-accenture-default          cert-tls-odbc-1-2883889434                valid
29d
```

If succesful the cert CRD should create a plain old k8s secret.

```
neil.toor@Toor-Neils-MacBook-Pro ~ % kubectl get secret -n beta3-dataviz-accenture-default
NAME                            TYPE                             DATA   AGE
cert-tls                        kubernetes.io/tls                2      79d
cert-tls-odbc                   kubernetes.io/tls                2      79d
```

Then ensure the secret is referenced correctly in the ingress.

```
kubectl edit ingress -n beta3-dataviz-accenture-default

#look for TLS section and SecretName should match the secret above.
  tls:
  - hosts:
    - dv-accenture-default-odbc.skylar-beta.sciencelogic.com
    secretName: cert-tls-odbc
```

Node management via Karpenter:

The logs are quite helpful and should be the first thing to check.

```
kubectl get pods -n karp
NAME                          READY   STATUS    RESTARTS   AGE
karpenter-69f9cbc7d6-fwmtc    1/1     Running   0          82d
karpenter-69f9cbc7d6-qgkj9    1/1     Running   0          82d

#Generally look for any exceptions in the logs
kubectl logs karpenter-69f9cbc7d6-fwmtc -n karp


#If you want to determine why Karpenter killed a node grep for reason.
kubectl logs karpenter-69f9cbc7d6-fwmtc -n karp  | grep -i "reason\":"
```

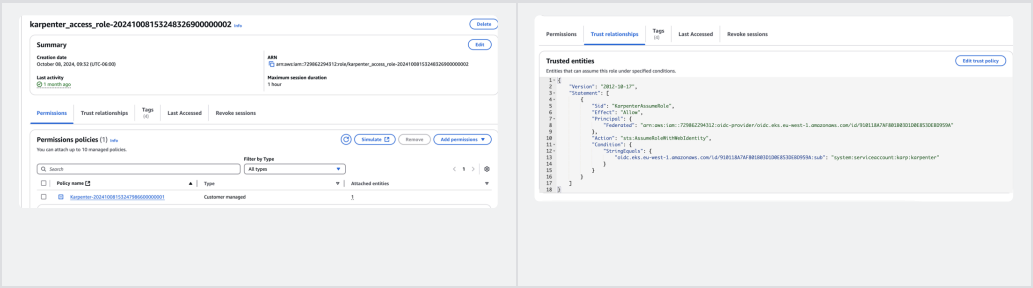
If logs show authentication issues to AWS then you can check IAM role mapping.

1. Find the SA that's linked to Karpenter.

```
kubectl get sa karpenter -n karp -o yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  annotations:
    eks.amazonaws.com/role-arn: arn:aws:iam::729862294312:role/karpenter_access_role-
20250115002632529800000008
```

2. Ensure SA is built in AWS

   The role should have a karpenter permissions policy (created by the Terraform) and correct Trust Relationship. Sample Screenshots below.



How to Check:

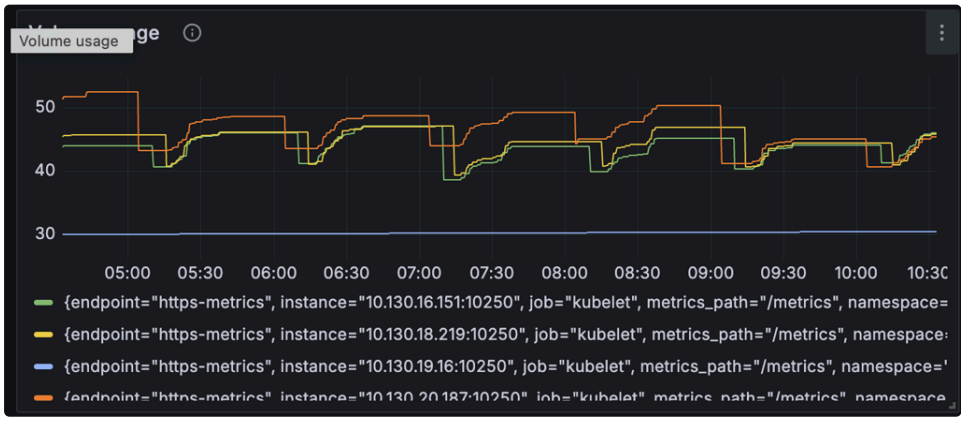
Disk Space capacity:

Check Grafana dashboard for Disk usage panel or use prometheus query.

Prometheus query:

```
kubelet_volume_stats_used_bytes / kubelet_volume_stats_capacity_bytes * 100 > 30
```

Grafana:



If you need to increase disk space you will need to update the inventory file and run the PVC increase steps

Skylar upgrade guide. | Increasing PVC size: